

$$|\Psi^-\rangle = (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)/\sqrt{2}$$

non-separable joint DI-bit state



Detector A
axis: 0° , outcome +1

$$E(\hat{a}, \hat{b}) = -\cos \theta$$

$\theta = 45^\circ \rightarrow -0.707$

Detector B
axis: 45° , outcome -1